

Does conditioning leave a structural memory?

Large-area structural evidence for the dislocation picture of conditioning

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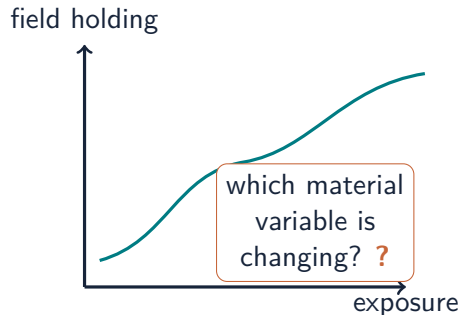
I. Popov (HUJI Nano) · V. M. Bjelland, W. Wuensch, L. Millar, S. Calatroni (CERN)



Conditioning works. What does it change?



- ▶ Conditioning is how every high-gradient device reaches its usable field.
- ▶ Breakdown rate has an extreme field dependence, $\sim E^{30}$.
- ▶ It tracks the number of **high-field pulses**, not the number of breakdowns.
- ▶ The field itself drives conditioning, not the arcs.



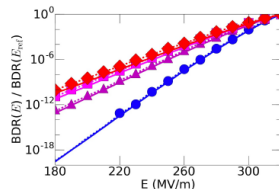
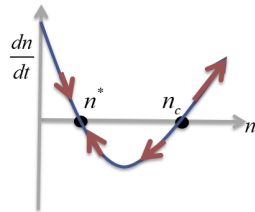
Something in the metal stores the improvement. We do not know what.

Source: Wuensch et al., Rep. Prog. Phys. (2026); Degiovanni et al., PRAB (2016).

Our model: conditioning as collective dislocation dynamics

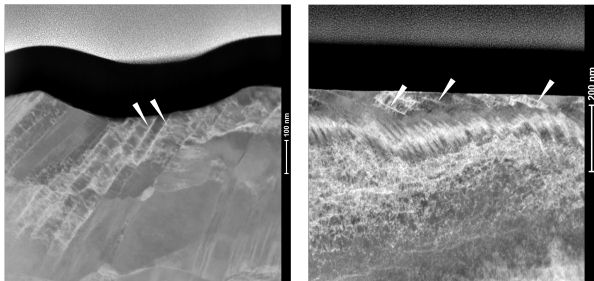


- ▶ Field holding rises FCC→BCC→HCP — the order of barriers to dislocation motion.
- ▶ The **MDDF** model: breakdown nucleates when the mobile-dislocation population, driven by Maxwell stress, runs away.
- ▶ One model gets the $\sim E^{30}$ rate, its temperature dependence, and the dark-current spikes.



Central prediction: conditioning *evolves the near-surface dislocation structure*.

STEM already saw it, at single spots



Cross-sectional STEM of conditioned Cu: dislocation-denuded zone in the top ~ 200 nm.

- ▶ First **direct** structural evidence of field-driven near-surface plasticity.
- ▶ Seen at both 300 K and 30 K.
- ▶ Fits the macroscopic clue: soft Cu conditions far more slowly than hard Cu, so the starting dislocation population matters.
- ▶ Hard lesson: FIB preparation plants its own dislocations. Trust only large-scale structure.
- ▶ One cross-section cannot speak for the whole electrode.

Confirmed at one spot. The large-area comparison was still missing.

A tiny stress, repeated a billion times



Maxwell stress

$$\sigma_M = \frac{\epsilon_0 E^2}{2}$$

at 80 MV/m: **0.028 MPa**

$\sim 10^3 \times$ below the yield of annealed Cu

- ▶ Fatigue at 10^8 – 10^{10} cycles rearranges dislocations irreversibly, well below yield.
- ▶ Here: $\sim 10^9$ conditioning pulses.
- ▶ Pulsed DC means **no RF pulsed heating** and no bulk currents. Maxwell stress is the only cyclic load.
- ▶ MDDF is **threshold-free**: no minimum stress amplitude.

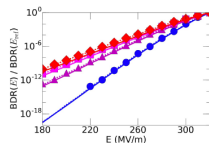
The model predicts a large-area signature. Last year we asked for it; this year we looked.

Source: Mughrabi, MMTB 40, 431 (2009); Stanzl-Tschegg et al., Int. J. Fatigue 29, 2050 (2007).

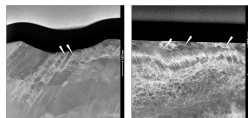
Last year we asked for more forensics. Here it is.



MeVArc 2025



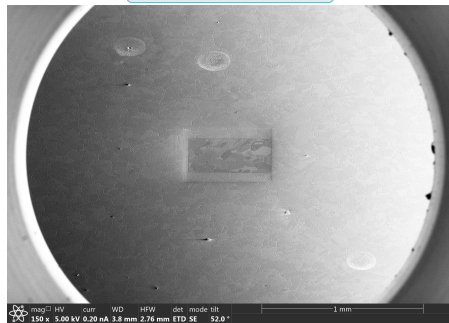
MDDF model



local STEM denuded
zone

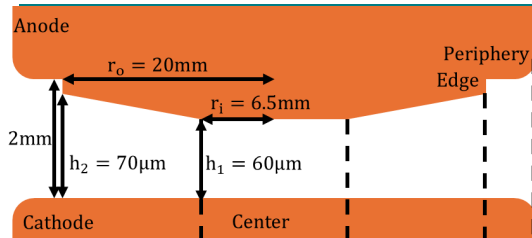
Closed with: other materials? more samples?

MeVArc 2026

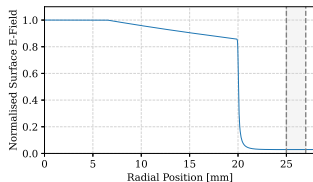


Large-area EBSD on one full cathode

One cathode turns field exposure into a controlled coordinate



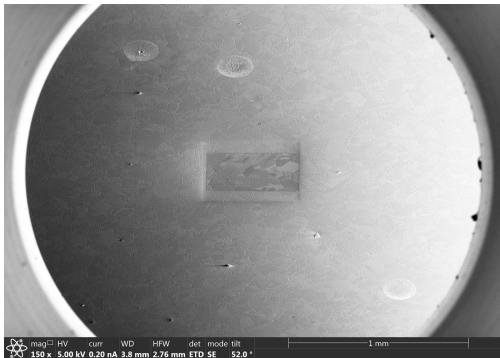
- ▶ OFE Cu, pulsed DC
- ▶ 1 μ s pulses at 1 kHz
- ▶ Peak field ~ 80 MV/m at center
- ▶ Nine EBSD regions: center, edge, periphery, and external reference



The field gradient gives us internal controls on a single specimen.

Source: Ashkenazy et al., submitted to PRAB; arXiv:2606.19192; geometry from Bjelland et al.

EBSD maps plastic activity far from breakdown craters



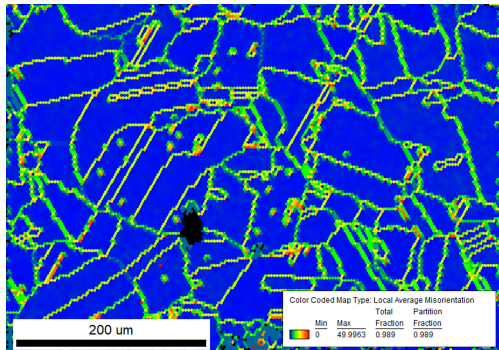
Cleaned ROI is hundreds of microns from visible craters.

- ▶ Large area where TEM could only sample a point
- ▶ 500 μm -wide maps; 3 μm step
- ▶ $\sim 24,000$ indexed points per ROI
- ▶ LAM, KAM, and LOS probe intragrain orientation gradients
- ▶ Misorientation is a proxy for GND density

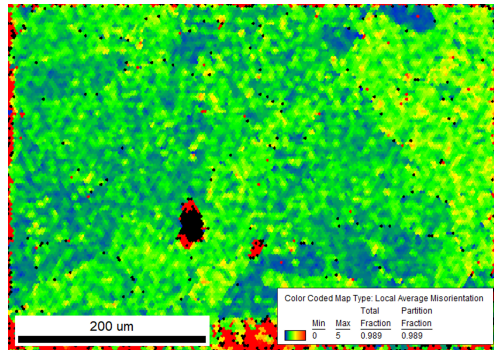
We measure the cumulative response to the field, not crater damage.

Source: Ashkenazy et al., submitted to PRAB; arXiv:2606.19192.

How to read a LAM map: wide-angle vs low-angle



0–50° scale: grain boundaries dominate

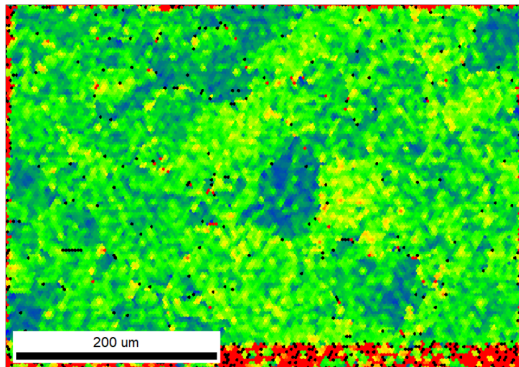


0–5° scale: intragrain curvature

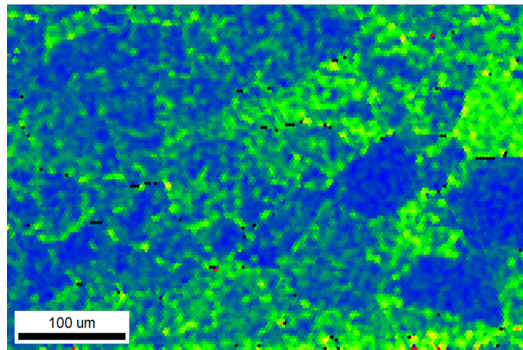
Rescaling to 0–5° exposes the subgrain misorientation we care about.

Source: Ashkenazy et al., submitted to PRAB; arXiv:2606.19192. Same area, two color scales.

High-field copper contains more intragrain curvature



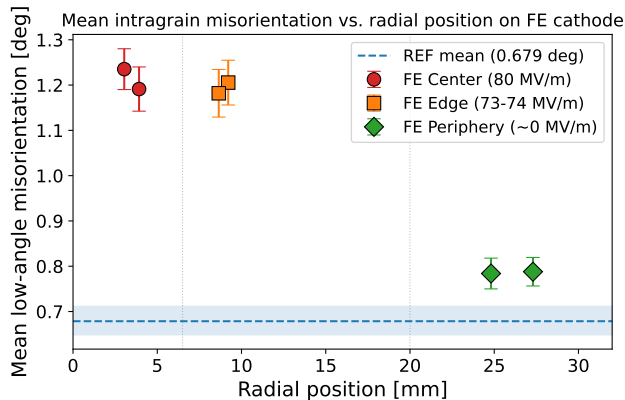
Field-exposed center



Unexposed reference

Source: Ashkenazy et al., submitted to PRAB; arXiv:2606.19192. Same LAM color scale.

The signal separates into three exposure tiers

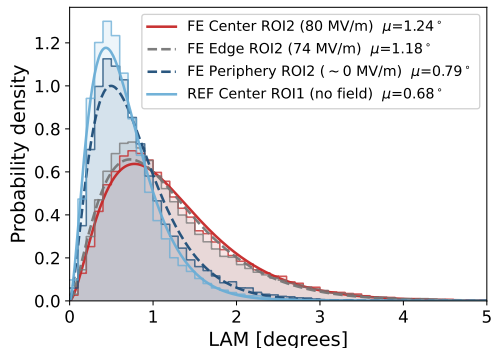


- Center and edge: $\sim 1.2^\circ$
- External reference: $\sim 0.68^\circ$
- High field is $\sim 75\%$ higher
- Grain size does not follow this hierarchy

The ordering follows field history, not sample position or grain size.

Source: Ashkenazy et al., submitted to PRAB; arXiv:2606.19192.

Field exposure builds a high-misorientation tail



LAM tail above 2°

~ 0.14 field-exposed

~ 0.016 reference

$\sim 8\times$ **increase**

Common gamma shape $k \approx 2.7$;

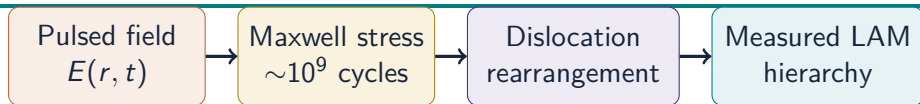
scale roughly doubles. All
center/edge vs unexposed:

$D_{KS} > 0.25$.

The whole distribution stretches; a uniform offset cannot do that.

Source: Ashkenazy et al., submitted to PRAB; arXiv:2606.19192.

The large-area test confirms the model's prediction



- ▶ First **large-area** confirmation that conditioning evolves the near-surface dislocation structure, as MDDF predicted.
- ▶ No conflict with the 2025 denuded zone: dislocations annihilate at the surface ($\ell_D \sim 100$ nm) and accumulate deeper down.
- ▶ The three-tier ordering also tracks the conditioning variable E_S : a *candidate* structural basis for it.

Before, dislocations were plausible. Now, a dislocation-sensitive state maps the conditioning history.

Source: Ashkenazy et al., submitted to PRAB; arXiv:2606.19192; Groma et al., PRL 96, 165503 (2006).

Copper establishes the effect, not universality



What is established

- ▶ Large-area structural difference
- ▶ Three independent metrics
- ▶ Internal and external controls
- ▶ Confirms the MDDF prediction

What remains open

- ▶ One conditioned cathode / one experiment
- ▶ Full depth profile
- ▶ Grain-orientation dependence
- ▶ More conditioned cathodes
- ▶ Cross-material generality

Call for cooperation: more cathodes, other materials, other facilities — let's compare notes.

A stronger test changes both the material and the loading history.

Next: go deeper, and change the material



Depth

- ▶ Matched high-field / reference pilot FIB lamellae
- ▶ Artifact controls learned from the 2025 “red herrings”
- ▶ Correlated TEM/STEM + EBSD through the top $\sim 1\ \mu\text{m}$
- ▶ Does the $\sim 200\ \text{nm}$ depletion join the deeper accumulation?

Depth continues the Uppsala STEM line; material tests the mechanism itself.

Material & loading

- ▶ Stainless steel: lower stacking-fault energy
- ▶ Different dislocation mobility and slip
- ▶ Continuous DC instead of pulsed loading
- ▶ Possible metastable phase transformation

Source: Interim work plan (2026); RFX project plan.

Stainless steel: a test chosen to be hard on the model



Copper

FCC, higher SFE
pulsed DC, $\sim 10^9$ cycles

AISI 304 SS

metastable austenite, lower SFE
continuous DC, possible $\gamma \rightarrow \alpha'$

SS tracks field

supports a cross-material structural state

SS differs

identifies material-specific carriers

no resolved change

bounds depth, dose, sensitivity

Design: apex-to-periphery gradient; two exposed + two matched references.

In collaboration with



CONSORZIO RFX
Ricerca Formazione Innovazione

Consorzio RFX — A. De Lorenzi, N. Pilan, E. Spada

Every outcome teaches us something. This is not a pass-or-fail test.

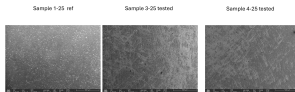
Source: RFX electrode geometry and plan; preliminary observations kept in backup.

RFX: first look (preliminary)

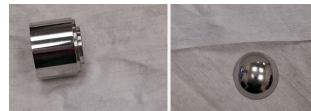
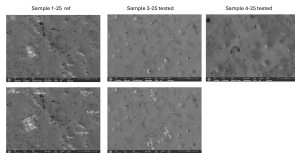


Preliminary — speaker may skip

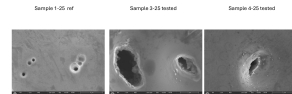
- ▶ AISI 304 SS at RFX
HVPTF; continuous DC,
 ≥ 100 h
- ▶ As-received SEM:
“exploded” near-surface
inclusions on field-exposed
samples
- ▶ Reference cavities round
and clean; morphology, not
size, differs
- ▶ First plan-view EBSD:
#1-25 good; #4-25 poor
patterns



Ref vs field-exposed overview



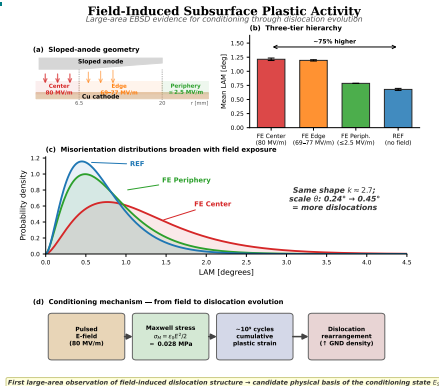
Sectioned electrode



Conditioning appears to leave a subsurface memory



1. Our model predicts conditioning evolves the near-surface dislocation structure.
2. 2025 STEM saw it locally; **large-area EBSD now confirms it** (~75% higher misorientation).
3. Depth and stainless steel test whether the memory survives a change of scale, alloy, and loading.



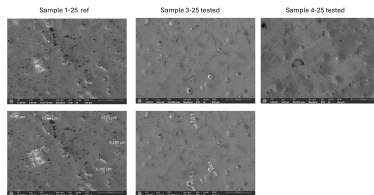
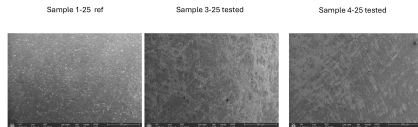
Last year: “more samples? other materials?” — this is the large-area answer.

Copper result: submitted to PRAB; arXiv:2606.19192.

Backup: RFX preliminary SEM survey



- ▶ No copper-like breakdown craters on as-received SS
- ▶ Field-exposed: “exploded” subsurface inclusions (3–7 μm)
- ▶ Reference (#1-25): round cavities, clean, inclusion-free
- ▶ EBSD: #1-25 good; #4-25 poor; #2-25/#3-25 follow-up pending



Source: Inna Popov, SEM report 20 May 2026; RFX status 13 Jul 2026.

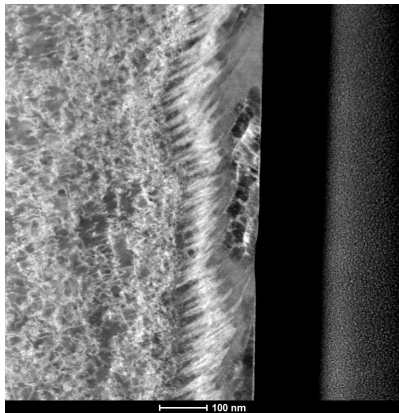
Backup: RFX sample and data status



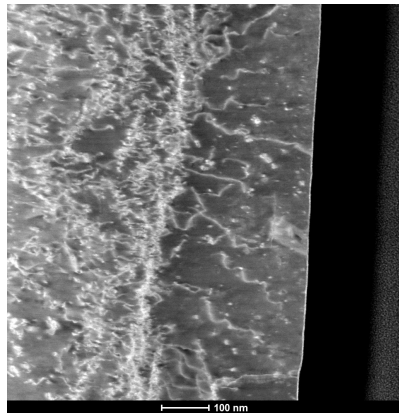
Sample	Role	Surface	Current status
#1-25	reference	electropolished	SEM + good EBSD
#2-25	reference	electropolished	follow-up acquisition planned
#3-25	field-exposed	≥ 100 h HV	SEM; EBSD follow-up
#4-25	field-exposed	≥ 100 h HV	SEM; poor initial EBSD

- ▶ Outstanding: full conditioning report; numerical $E(\theta)$ vector; alloy certificate; electropolishing protocol.
- ▶ Needed for interpretation: quantitative EBSD quality metrics and raw patterns; matched EDS overlay checks.

Backup: field-exposed vs reference STEM (2025)



Field-exposed — denuded near surface



Reference — dislocation-rich to surface

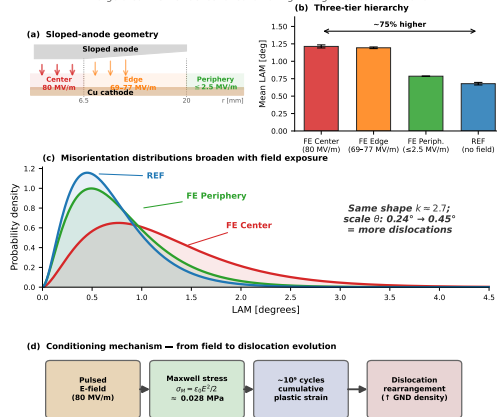
Source: Jacewicz et al., JAP 137, 193302 (2025).

Backup: copper result in one figure



Field-Induced Subsurface Plastic Activity

Large-area EBSD evidence for conditioning through dislocation evolution



First large-area observation of field-induced dislocation structure → candidate physical basis of the conditioning state E_s